Decision Tree Algorithm.

With default parameter (*squared error) R Value= 0.6831243367079769*
https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html

S.no	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1.	squared error	*sqrt*	best	0.69045480031074
2	friedman_mse	sqrt	best	0.6642036336908992
3	friedman_mse	sqrt	random	0.7064353558774805
4	friedman_mse	log2	best	0.6846491435674211
5	friedman_mse	log2	random	0.7389567370162474
6	poisson	log2	best	0.6950142840696334
7	poisson	log2	random	0.6832926465346087
8	poisson	sqrt	best	0.7129599184895257
9	poisson	sqrt	random	0.7168468995276225
10	absolute_error	sqrt	random	0.6141007044431557
11	absolute_error	Log2	random	0.6709031189911441
12	absolute_error	sqrt	best	0.7963292022757553
13	absolute_error	Log2	best	0.7096641680204572
14	squared error	sqrt	random	0.70508686930045
15	squared error	log2	random	0.7103743484511277
16	squared error	log2	best	0.7025538304159292

SVM

http://scikit-learn.org/stable/modules/generated/sklearn.svm.SVR.html

Kernal =poly - > -0.07970221554339751

S.no	Parameters	R Value
1.	**Kernal =poly**	**-0.07970221554339751**
2	**Kernal =linear**	**-0.14846453125202963**
3	***Kernal=sigmoid***	
4	**SVR(kernel='rbf', C=1.0, epsilon=0.1, gamma='scale', degree=3, coef0=0.0, shrinking=True, tol=0.001, cache_size=200, verbose=False, max_iter=-1**	**-0.0979940352451043**
5	***precomputed***	**ValueError: Precomputed matrix must be a square matrix. Input is a 892x5 matrix. (Note: Our dataset is different)**

Random Forest

S.no	Parameters	R Value
1.	regressor=RandomForestRegressor(n_estimators =100,random_state=0) // default	0.857655311493 1038
2	regressor=RandomForestRegressor(n_estimators=100,max_depth=8)	0.872872020765 9969
3	regressor=RandomForestRegressor(n_estimators=100,max_depth=10,r andom_state=42)	0.864135761301 8931
4	RandomForestRegressor(n_estimators=200, criterion="squared_error", max_depth=15, min_samples_split=5, random_state=0) regressor=regressor.fit(X_train,y_train)	0.870713944102 2052
5	regressor = RandomForestRegressor(n_estimators=150, criterion="squared_error", max_depth=12, min_samples_split=4, min_samples_leaf=2, max_features="sqrt", random_state=0)	0.882125537449 8428
6	regressor = RandomForestRegressor(n_estimators=300, criterion="squared_error", max_depth=20, min_samples_split=5, min_samples_leaf=2, max_features="sqrt", bootstrap=True, oob_score=True, n_jobs=-1, random_state=42)	0.880752914616 5676
7	from sklearn.ensemble import RandomForestRegressor regressor = RandomForestRegressor(n_estimators=200, max_depth=15, min_samples_split=5, min_samples_leaf=2, max_features="sqrt", bootstrap=True, random_state=42, n_jobs=-1)	0.880828667075 8186
8	regressor = RandomForestRegressor(n_estimators=200, max_depth=None, min_samples_split=5, min_samples_leaf=2, max_features="sqrt",	0.881040354387 8095

	bootstrap=True, **random_state=42,** **n_jobs=-1** **)**	